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creates
manages
Extends
<<uses>>
Text
User
+userID: int
+name: String
+email: String
+role: String
+login() : : boolean
+logout() : : void
+viewDashboard() : : void
SuspiciousActivity
- reportID
- transactionID
- reason
- timestamp
+ saveReport(): void
CashDrawer
- systemTotal
- actualTotal
- difference
+ compareCash(actual): float
+ checkDifference(): bool
ReturnRecord
- returnID
- productID
- transactionID
- refundAmount
+ verifyEligibility(): bool
+ processRefund(): float
SalesReport
- date
- totalTransactions
- grossSales
- netSales
+ generateReport(): void
+ summarizeTotals(): void
Cashier
- cashierID
- name
- shiftEnd
+ startCheckout(): void
+ enterProductID(id): void
+ applyMembership(id): void
+ processPayment(method, details): void
+ handleReturn(productID, billNo): void
+ generateSalesReport(): void
+ balanceCashDrawer(actualCash: float): void
+ nextCustomer(): void
+ flagSuspicious(txnID, reason): void
+ displayBill(): void
+ displayStatus(): void
Use
Transaction
- transactionID
- itemList: List<Item>
- subtotal
- discount
- totalAmount
- paymentStatus
- dateTime
+ addItem(item: Item): void
+ calculateTotal(): float
+ applyMemberDiscount(rate: float): void
+ validatePayment(): bool
+ processRefund(): float
+ summarize(): ReportData
Item
- itemID
- itemName
- unitPrice
- quantity
+ getPrice(): float
Membership
- memberID
- memberName
- discountRate
- isValid
+ verifyMembership(id): bool
+ getRate(): float
PaymentRecord
- paymentID
- transactionID
- method
- amount
- status
+ validatePayment(): bool
+ savePayment(): void
InventoryManager
- managerID
- name
+ checkInventory(): void
+ raisePurchaseRequest(pid, qty): void
+ receiveDelivery(details): void
+ recordDamagedGoods(pid, qty): void
+ generateForecast(): void
+ generateStockReport(): void
+ assignStorageLocation(pid, location): void
+ returnToSupplier(pid, qty): void
+ displayResult(): void
Inventory
- productList: List<Product>
+ fetchStockData(): List<Product>
+ updateStock(pid, qty): void
+ getProduct(pid): Product
DamageRecord
- damageID
- productID
- quantity
- date
+ verify(): bool
+ deductStock(): void
ForecastEngine
- salesHistory
- forecastData
+ fetchSalesHistory(): void
+ calculateForecast(): void
Product
- productID
- productName
- category
- quantityInStock
- location
+ adjustStock(qty): void
+ setLocation(loc): void
SupplierDelivery
- invoiceNo
- deliveredItems: List<DeliveryItem>
+ verifyWithPO(): bool
+ updateInventory(): void
StockReport
- reportDate
- lowStockItems
- overStockItems
- fullInventory
+ generateReport(): void
+ processTotals(): void
PurchaseRequest
- requestID
- productID
- quantity
- status
+ validateRequest(): bool
+ saveRequest(): void
StorageLocation
- shelfID
- section
- isAvailable
+ validateLocation(): bool
+ assignProduct(): void
SupplierReturn
- returnID
- productID
- quantity
- status
+ verifyReturn(): bool
+ processReturn(): void
parent
child
DeliveryCoordinator
+coordinatorID: int
+name: String
+department: String
+assignDeliveryTasks() : : boolean
+trackDelivery() : : String
+confirmCompletedDeliveries() : : boolean
+reportReturnedOrderProblem() : : boolean
+manageDeliverySchedule() : : boolean
+generateDeliveryReport() : : boolean
+verifyReturnedOrders() : : boolean
+processReDeliveryRequest() : : boolean
confirms >
tracks >
assigns >
verifies >
generates >
uses >
creates >
processes >
reports >
checks >
manages >
may become >
has >
may have >
ConfirmCompletedDelivery
+deliveryID: int
+orderID: int
+completionDate: LocalDate
+staffID: int
+confirmDelivery() : : boolean
+validateOrder() : : boolean
AssignDeliveryTasks
+orderID: int
+staffID: int
+deliveryDate: LocalDate
+status: String
+assignTask() : : boolean
+validateInputs() : : boolean
+saveRecord() : : boolean
VerifyReturnedOrder
+returnID: int
+orderID: int
+returnReason: String
+status: String
+verifyReturn() : : boolean
+filterProblems() : : List
+displayProblems() : : void
GenerateDeliveryReport
+reportID: int
+date: LocalDate
+totalOrders: int
+deliveredOrders: int
+pendingOrders: int
+generateReport() : : boolean
+downloadPDF() : : boolean
+displayReport() : : void
ReDeliveryRequest
+requestID: int
+orderID: int
+reason: String
+requestDate: LocalDate
+status: String
+saveRequest() : : boolean
+verifyOrderStatus() : : boolean
+removeFromConfirmedOrders() : : boolean
Track Delivery
+statusID: int
+orderID: int
+currentStatus: String
+lastUpdated: LocalDateTime
+checkStatus() : : String
+validateOrderID() : : boolean
ManageDeliverySchedule
+scheduleID: int
+orderID: int
+deliveryDate: LocalDate
+previousDate: LocalDate
+updateSchedule() : : boolean
+validateOrder() : : boolean
+saveNewDate() : : boolean
ReportReturnedOrderProblem
+problemID: int
+orderID: int
+problemDetails: String
+reportDate: LocalDate
+status: String
+reportProblem() : : boolean
+validateInfo() : : boolean
+verifyDeliveryStatus() : : boolean
ProcurementOfficer
+officerID: int
+name: String
+department: String
+reviewPurchaseRequest() : : void
+createSupplierOrder() : : boolean
+approveSupplierQuotation() : : boolean
+resolveIssuesWithSupplier() : : boolean
+manageSupplierRecords() : : boolean
+trackPendingSupplierDeliveries() : : void
+generateProcurementReport() : : boolean
+viewSupplierContactInformation() : : void
approves >
reviews >
is tracked as >
may generate >
resolves >
approves/rejects >
views >
ManageSupplierRecord
+supplierID: int
+name: String
+contact: String
+address: String
+saveRecord() : : boolean
+validateInputs() : : boolean
+updateRecord() : : boolean
ReviewPurchaseRequest
+requestID: int
+productID: int
+quantity: int
+status: String
+showRequests() : : void
+fetchFromManager() : : void
+displayRequests() : : void
CreateSupplierOrder
+orderID: int
+supplierID: int
+product: String
+quantity: int
+dueDate: LocalDate
+status: String
+saveOrder() : : boolean
+validateInputs() : : boolean
+generateOrderID() : : String
View Supplier Contact Information
+supplierID: int
+name: String
+contact: String
+address: String
+searchSupplier() : : void
+displayDetails() : : void
+validateSupplierID() : : boolean
GenerateProcurementReport
+reportID: int
+date: LocalDate
+orders: List
+totalCost: Double
+generateReport() : : boolean
+downloadPDF() : : boolean
+displayOrders() : : void
ApproveSupplierQuotation
+quotationID: int
+supplierID: int
+details: String
+price: Double
+status: String
+approveQuotation() : : boolean
+rejectQuotation() : : boolean
+saveDecision() : : boolean
TrackPendingDeliveries
+deliveryID: int
+orderID: int
+supplierID: int
+dueDate: LocalDate
+status: String
+loadOrders() : : void
+loadDeliveryStatus() : : void
+compareDeliveries() : : void
+displayPendingOrders() : : void
ResolveSupplierIssue
+issueID: int
+supplierID: int
+damagedProduct: String
+remarks: String
+status: String
+status: String
+loadDamagedProducts() : : void
+sendReplacementRequest() : : boolean
+displayProducts() : : void
StoreManager
+managerID: int
+name: String
+department: String
+overseeDailyOperations() : : boolean
+approvePurchaseOrder() : : boolean
+assignStaffShifts() : : boolean
+monitorSalesPerformance() : : void
+handleStaffComplaints() : : boolean
+approveDiscountsAndPromotions() : : boolean
+generateTopManagementReports() : : void
+assignStaffTask() : : boolean
approves/rejects >
monitors >
handles >
DailyOperation
+operationID: int
+date: LocalDate
+totalSales: Double
+lowStockInfo: String
+activeStaffCount: int
+saveOperation() : : boolean
+displayConfirmation() : : void
PurchaseRequest
+requestID: int
+productID: int
+productDetails: String
+quantity: int
+status: String
+approve() : : boolean
+reject() : : boolean
StaffShift
+shiftID: int
+staffID: int
+shiftDate: LocalDate
+shiftTime: String
+assignShift() : : boolean
+saveToDatabase() : : boolean
SalesData
+dataID: int
+date: LocalDate
+totalSales: Double
+transactionCount: int
+loadFromCashierFiles() : : void
+displayReport() : : void
StaffComplaint
+complaintID: int
+staffID: int
+complaintDetails: String
+date: LocalDate
+status: String
+saveComplaint() : : boolean
+displayConfirmation() : : void
Promotion
+promotionID: int
+productTitle: String
+previousPrice: Double
+promoPrice: Double
+validFrom: LocalDate
+validTo: LocalDate
+savePromotion() : : boolean
+validateBudget() : : boolean
ManagementReport
+reportID: int
+date: LocalDate
+dailyOperationRecords: String
+summary: String
+generateReport() : : void
+displayReport() : : void
StaffTask
+taskID: int
+staffID: int
+taskDate: LocalDate
+taskArea: String
+assignTask() : : boolean
+saveAssignment() : : boolean
MembershipManager
+managerID: int
+name: String
+department: String
+registerNewMembership() : : boolean
+renewMembership() : : boolean
+cancelMembership() : : boolean
+handleLostCardRequests() : : boolean
+updateMembershipBenefits() : : boolean
+generateMembershipReports() : : void
+viewMembershipInfo() : : Membership
+handleMembershipIssues() : : boolean
resolves >
updates >
entitled to >
MembershipReport
+reportID: int
+date: LocalDate
+membershipRecords: List
+generatePDF() : : boolean
+displayRecords() : : void
MembershipIssue
+issueID: int
+membershipID: int
+issueDescription: String
+status: String
+loadIssue() : : void
+displayIssue() : : void
Membership
+membershipID: int
+memberName: String
+membershipType: String
+validFrom: LocalDate
+validTo: LocalDate
+status: String
+generateID() : : String
+saveRecord() : : boolean
+extendValidity() : : boolean
+cancel() : : boolean
MembershipCard
+cardID: int
+membershipID: int
+issueDate: LocalDate
+status: String
+generateNewCard() : : boolean
+saveUpdatedID() : : boolean
MembershipBenefit
+benefitID: int
+membershipID: int
+benefitDetails: String
+validFrom: LocalDate
+validTo: LocalDate
+updateBenefit() : : boolean
+saveInformation() : : boolean
CustomerSupportStaff
- supportID: int {unique}
- name: String
- department: String
- shift: String
- accessLevel: String
+ addComplaint(): boolean
+ updateComplaint(): boolean
+ closeComplaint(): boolean
+ checkOrderStatus(): String
+ registerMembershipRequest(): boolean
+ viewPendingComplaints(): List<Complaint>
+ submitCustomerFeedback(): boolean
+ generateSupportReport(): String
+ loadPendingComplaints(): List<Complaint>
parentrchComplaintsByCustomer(): List<Complaint>
+ validateComplaintData(): boolean
Customer Support - ALL 8 Goals:
1. Add New Complaint
2. Update Complaint Info
3. Close Complaint
4. Check Order Status
5. Register Membership Request
6. Pending Complaint Reminder
7. Submit Customer Feedback
8. Generate Support Report
HR Manager
- hrID: int {unique}
- name: String
- department: String
- accessLevel: String
+ recruitStaff(): boolean
+ removeEmployee(): boolean
+ generateHRReport(): String
+ manageAttendance(): boolean
+ managePayroll(): boolean
+ submitLeaveRequest(): boolean
+ viewHRSummaryReport(): String
+ exportToPDF(): boolean
+ loadEmployees(): List<Employee>
+ saveToFile(): boolean
+ validateEmployeeData(): boolean
+ clearAttendanceTable(): void
HR Manager - ALL 8 Goals:
1. Recruit Staff
2. Remove Employee
3. Generate HR Report
4. (Reserved)
5. Manage Attendance
6. Manage Payroll
7. Leave Request
8. View HR Summary Report
Recruitment
- recruitmentID: int
- employeeID: int
- recruitedBy: int
- recruitmentDate: LocalDate
- personalDetails: String
- jobDetails: String
- department: String
- position: String
+ saveStaffData(): boolean
+ validateRequiredFields(): boolean
+ displayConfirmation(): void
+ loadDepartments(): List<String>
+ loadPositions(): List<String>
+ generateUniqueID(): int
Employee Removal
- removalID: int
- employeeID: int
- removedBy: int
- removalDate: LocalDate
- reason: String
+ verifyEmployeeID(): boolean
+ deleteEmployeeData(): boolean
+ displayConfirmation(): void
+ loadEmployeeTable(): void
+ updateEmployeeList(): boolean
Employee
- employeeID: int {unique}
- name: String
- phone: String
- salary: double
- position: String
- department: String
- joiningDate: LocalDate
- status: String
+ addEmployee(): boolean
+ removeEmployee(): boolean
+ updateInfo(): boolean
+ validateFields(): boolean
+ generateUniqueID(): int
+ updateSalary(newSalary: double): boolean
Update Employee Info
- updateID: int
- employeeID: int
- updatedBy: int
- updateDate: LocalDate
- fieldChanged: String
- oldValue: String
- newValue: String
+ fetchEmployeeData(): Employee
+ editDetails(): boolean
+ saveUpdatedInfo(): boolean
+ showConfirmation(): void
View HR Summary Report
- reportID: int
- generatedBy: int
- generationDate: LocalDate
- reportData: String
+ loadEmployeeData(): List<Employee>
+ generateReportContent(): String
+ exportToPDF(): boolean
+ validateReportData(): boolean
+ displayReport(): void
Approve Leave Requests
- requestID: int
- employeeName: String
- leaveType: String
- startDate: LocalDate
- endDate: LocalDate
- status: String
- submittedBy: int
+ submitRequest(): boolean
+ loadRequests(): List<LeaveRequest>
+ validateFields(): boolean
+ saveToFile(): boolean
+ clearTable(): void
+ clearInputFields(): void
Manage Staff Attendance
- attendanceID: int
- employeeID: int
- employeeName: String
- date: LocalDate
- status: String
+ markAttendance(): boolean
+ loadEmployees(): List<Employee>
+ saveAttendance(): boolean
+ clearAttendanceTable(): void
+ validateAttendanceData(): boolean
Leave Request
- requestID: int
- employeeName: String
- leaveType: String
- startDate: LocalDate
- endDate: LocalDate
- status: String
- submittedBy: int
+ submitRequest(): boolean
+ loadRequests(): List<LeaveRequest>
+ validateFields(): boolean
+ saveToFile(): boolean
+ clearTable(): void
+ clearInputFields(): void
Manage Payroll
- payrollID: int
- employeeID: int
- currentSalary: double
- newSalary: double
- updateDate: LocalDate
- updatedBy: int
+ loadEmployees(): List<Employee>
+ updateSalary(): boolean
+ validateSalaryInput(): boolean
+ saveToFile(): boolean
+ displaySelectedEmployeeSalary(): void
Complaint
- complaintID: int {unique}
- customerID: int
- customerName: String
- complaintTitle: String
- description: String
- status: String
- dateReported: LocalDate
- resolvedDate: LocalDate
- assignedTo: int
+ addComplaint(): boolean
+ updateComplaint(): boolean
+ closeComplaint(): boolean
+ markAsSolved(): boolean
+ validateFields(): boolean
+ fetchComplaintData(): Complaint
+ saveToFile(): boolean
+ loadFromFile(): List<Complaint>
+ filterPendingComplaints(): List<Complaint>
+ searchByCustomerName(): List<Complaint>
Check Customer Order Status
- orderID: int {unique}
- customerID: int
- orderDate: LocalDate
- status: String
- deliveryStatus: String
- totalAmount: double
- progress: String
+ checkOrderStatus(): String
+ updateOrderProgress(): boolean
+ validateOrderID(): boolean
+ fetchOrderDetails(): Order
+ displayOrderProgress(): void
Membership
- membershipID: int {unique}
- customerID: int
- membershipType: String
- startDate: LocalDate
- expiryDate: LocalDate
- status: String
+ registerMembership(): boolean
+ validateRequiredInfo(): boolean
+ saveRecord(): boolean
+ displayConfirmation(): void
Membership Request
- requestID: int {unique}
- customerName: String
- contactInfo: String
- notes: String
- status: String
- requestDate: LocalDate
+ submitRequest(): boolean
+ validateRequiredFields(): boolean
+ saveToFile(): boolean
+ clearForm(): void
+ displayConfirmation(): void
Customer Feedback
- feedbackID: int
- customerName: String
- feedbackContent: String
- submissionDate: LocalDate
- submittedBy: int
+ submitFeedback(): boolean
+ validateFields(): boolean
+ saveToFile(): boolean
+ clearInputFields(): void
+ displayConfirmation(): void
Generate Support Report
- reportID: int
- customerName: String
- status: String
- generatedBy: int
- reportDate: LocalDate
+ loadAllComplaints(): List<Complaint>
+ generateReport(): String
+ displayInTable(): void
+ validateReportData(): boolean
+ exportReport(): boolean
Pending Complaint Reminder
- reminderID: int
- complaintCount: int
- lastUpdated: LocalDate
+ loadPendingComplaints(): List<Complaint>
+ filterNonClosedComplaints(): List<Complaint>
+ displayPendingTable(): void
+ refreshReminder(): void
Customer
- customerID: int {unique}
- name: String
- phone: String
- email: String
- address: String
- membershipID: int
+ registerCustomer(): boolean
+ updateInfo(): boolean
+ viewOrders(): List<Order>
+ validateContactInfo(): boolean
File Management
- fileName: String
+ readEmployeeFile(): List<Employee>
+ writeEmployeeFile(employees: List<Employee>): boolean
+ readComplaintFile(): List<Complaint>
+ writeComplaintFile(complaints: List<Complaint>): boolean
+ readAttendanceFile(): List<Attendance>
+ writeAttendanceFile(attendance: List<Attendance>): boolean
+ readLeaveRequestFile(): List<LeaveRequest>
+ writeLeaveRequestFile(requests: List<LeaveRequest>): boolean
+ readMembershipRequestFile(): List<MembershipRequest>
+ writeMembershipRequestFile(requests: List<MembershipRequest>): boolean
+ readFeedbackFile(): List<CustomerFeedback>
+ writeFeedbackFile(feedback: List<CustomerFeedback>): boolean
File Operations:
- employee.bin
- complaints.txt
- attendance.bin
- leaveRequests.bin
- membership_requests.txt
- feedback.txt
1
*
0..*
0..1
child